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ME193: Computational Fluid Dynamics
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Numerical Simulation of Two-Dimensional Lid-Driven Cavity Flow Using COMSOL Multiphysics

I. Introduction

The lid-driven cavity problem is a widely studied problem in computational fluid dynamics (CFD) owing to its dynamic flow physics and significant dependence on the Reynolds number. The geometry is very simple and consists of a square cavity in which flow is driven by the tangential motion of the top wall, while all other walls remain stationary with no-slip conditions placed on them. The problem is ideal for studying the transition from viscous- to inertia-dominated flow regimes.

At low Reynolds numbers, viscous forces dominate and the flow is characterized by a single smooth primary vortex centered near the middle of the cavity. However, as the Reynolds number increases, inertial effects become more significant, causing the primary vortex to shift toward the right of the cavity and secondary vortices to develop in the corners. This sensitivity to the Reynolds number makes the lid-driven cavity an interesting problem for understanding how flow structure changes with changing flow conditions.

In this study, the two-dimensional incompressible Navier-Stokes equations are solved numerically for two Reynolds numbers cases, Re = 1 and Re = 500, using COMSOL Multiphysics®. The dimensionless governing equations the solver used are given by:

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u}\mathbf{u}) = -\nabla p + \frac{1}{Re}\nabla^2 \mathbf{u}$$

$$\nabla \cdot \mathbf{u} = 0$$

where **u** is the velocity vector, p is the pressure, and Re is the Reynolds number. The first equation represents momentum conservation, in which the term $\partial \mathbf{u}/\partial t$ represents the local acceleration of the fluid, while $\nabla \cdot (\mathbf{uu})$ is the convective term accounting for momentum transport due to the motion of the fluid itself. Together these two terms describe the total rate of change of momentum following the flow. On the right-hand side, $-\nabla p$ is the pressure gradient, describing the force driving fluid from high- to low-pressure regions, and $(1/Re)\nabla^2 u$ is the viscous diffusion term. This term represents the tendency of viscosity to smooth out velocity differences within the fluid by diffusing momentum from regions of high velocity to regions of low velocity, thus describing how much the fluid resists deformation due to internal friction. The ratio 1/Re controls the relative importance of viscous forces; at low Reynolds numbers this term is large and viscosity dominates, producing smooth flows, while at high Reynolds numbers it becomes small and the inertial effects of convective term take over, leading to more complex flow structures. The continuity equation $\nabla \cdot \mathbf{u} = 0$ enforces the incompressibility constraint, ensuring that mass is conserved and that there is no net compression or expansion of the fluid anywhere in the domain.

For each case of the Reynolds number, the velocity field, streamlines, vorticity distribution, and centerline velocity profiles are examined and compared to determine the effect of the Reynolds number on the cavity flow structure.

II. Methodology

2.1 Governing Equations

The flow is governed by the dimensionless incompressible Navier-Stokes equations, including the momentum and continuity equations described in Section 1. The dimensionless form of these equations is obtained by scaling velocity by the lid velocity U, length by the cavity side length L, and pressure by ρU^2. Since U = L = 1 and density $\rho = 1$ in the dimensionless system, the Reynolds number is defined as $Re = UL/\nu = 1/\nu$, where ν is the kinematic viscosity (and the dynamic viscosity, $\mu = \nu\varrho$). Thus, all simulations can be fully characterized by the Reynolds number alone.

2.2 Computational Domain and Boundary Conditions

The domain for the simulation is a square cavity of dimensionless side length L = 1. The boundary conditions include a top wall, the driven lid, which moves in the positive x-direction with a dimensionless velocity u = 1, v = 0. All remaining walls (the bottom, left, and right) are assigned no-slip conditions where u = v = 0. A pressure point constraint of p = 0 was applied at the bottom left corner of the domain to give the pressure field a fixed reference value. Without this, the solver has no way to determine the absolute pressure level in a fully enclosed cavity and fails to converge.

2.3 COMSOL Setup and Solver Configuration

The simulations for this experiment were performed using the Laminar Flow (spf) physics module in COMSOL Multiphysics, which solves the incompressible Navier-Stokes equations under steady-state conditions. Since both Reynolds numbers are much less than 2000, a Laminar assumption for this internal flow is accurate for both cases. A stationary solver was used for both Reynolds numbers, as the goal of this study is to analyze the fully developed steady-state flow structure rather than the time-dependent evolution of the flow. Since COMSOL requires dimensional inputs, the dimensionless system was reproduced by setting the fluid density to $\rho = 1$ kg/m³ and the lid velocity to U = 1 m/s over a domain of L = 1 m, such that the dynamic viscosity $\mu = 1/Re$ directly enforces the desired Reynolds number. For Re = 1, $\mu = 1.0$ Pa·s was used, and for Re = 500, $\mu = 0.002$ Pa·s was used. Two separate studies were configured within the same COMSOL project, one for each Reynolds number case, allowing the application of independent mesh and solver settings to each simulation.

2.4 Mesh Discretization

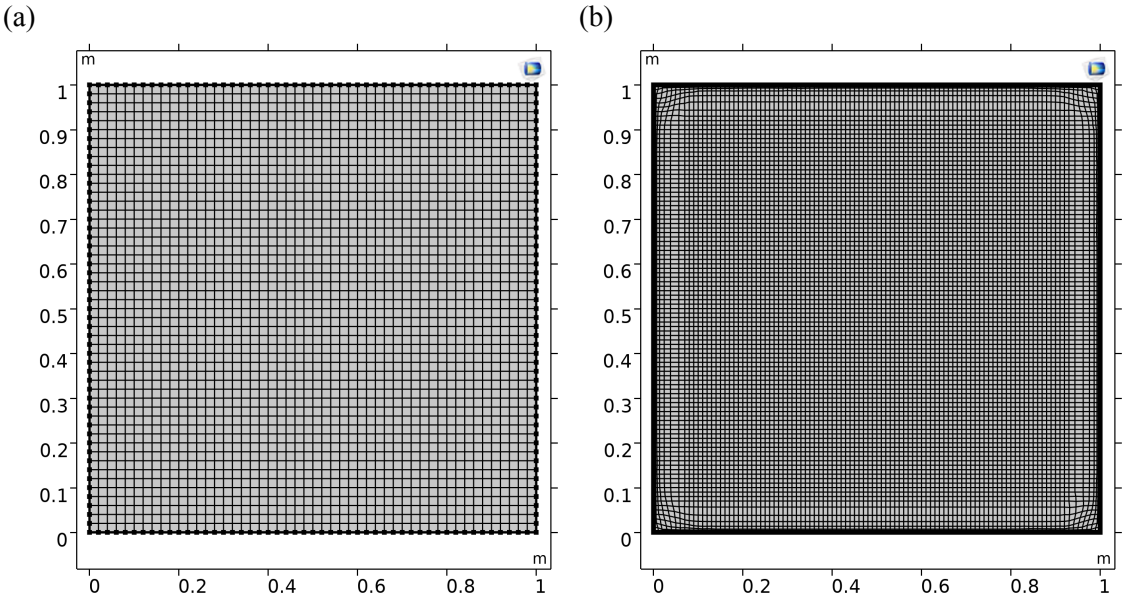


Fig 1. Mapped mesh discretizations for the (a) Re = 1 case and (b) Re = 500 case.

The domain was discretized using a mapped mesh with quadrilateral elements, which is well-suited for the square cavity geometry since it consists of uniformly shaped cells aligning with the domain boundaries. For the Re = 1 case, a uniform 50×50 element mesh was used, since the low Reynolds number produces smooth velocity gradients throughout the domain that do not require high spatial resolution to maintain accuracy. In this case, viscous forces dominate and velocity gradients are gentle and smoothly distributed throughout the entire cavity, making uniform meshing sufficient.
For the Re = 500 case, the mesh was refined to a 100×100 element base mesh to address the steeper velocity gradients and thinner boundary layers that develop at higher Reynolds numbers. Additional boundary layer refinement was applied along all walls for the Re = 500 simulation, which consisted of 4 layers with a thickness adjustment factor of 1.2 and the default stretching factor of 1.2, ensuring sufficient resolution for flow features near the walls, where velocity gradients steep. In higher Reynolds number cases like these, inertial forces become much more significant relative to viscous forces, resulting in the thin boundary layers along the walls where velocity changes rapidly over a short distance. By applying additional refinement at the walls for the Re = 500 case, the boundary layer physics are captured without unnecessarily refining the entire domain, which requires excessive computational power. The mesh discretizations for both Reynolds number cases are shown in Figure 1.

III. Results and Discussion

3.1 Low Reynolds Number (Re = 1)

3.1.1 Velocity Magnitude and Vorticity Field

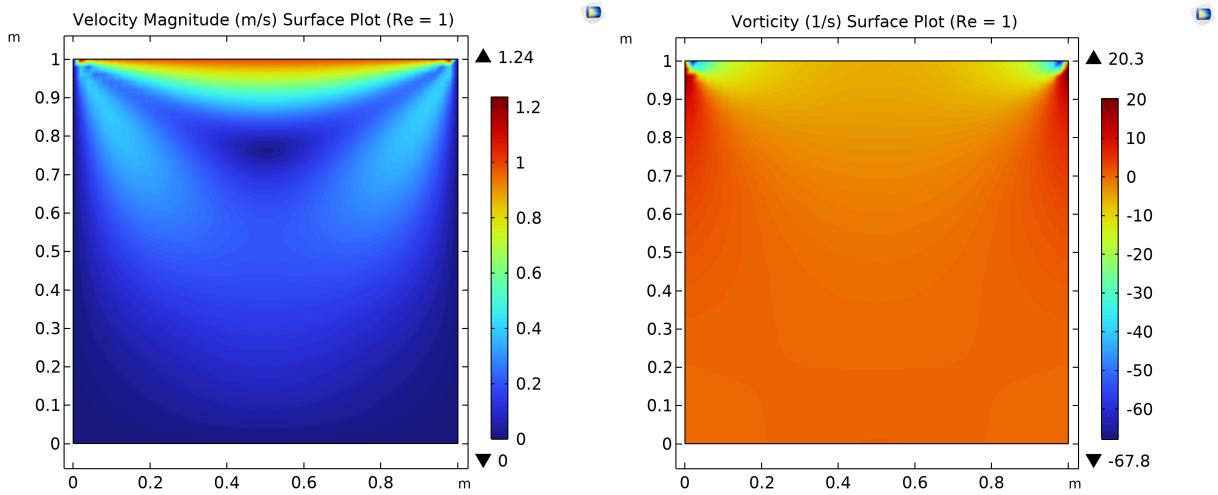


Fig 2. Velocity magnitude surface plot and vorticity surface plot for Re = 1 (side by side)

The velocity magnitude surface plot for Re = 1, shown in Figure 2, reveals that high velocity is concentrated almost entirely along the top lid, where the moving wall directly transfers momentum to the fluid. The interior of the cavity remains largely stagnant, with near-zero velocity magnitude throughout the remainder of the domain. This makes sense for the Re = 1 case since the flow is extremely viscous and viscous diffusion quickly dissipates momentum away from the lid before it can affect the fluid in the cavity's interior. The vorticity field, also shown in Figure 2, is negative throughout most of the domain, indicating a primarily clockwise rotational flow driven by the rightward-moving lid. Vorticity is most severe near the moving lid where the velocity gradient, $\partial u/\partial y$, is the largest, and decays smoothly toward the cavity interior and lower walls, reflecting the effects of viscous diffusion in spreading and smoothing the rotational effects throughout the domain.

3.1.2 Velocity Vector Field and Streamlines

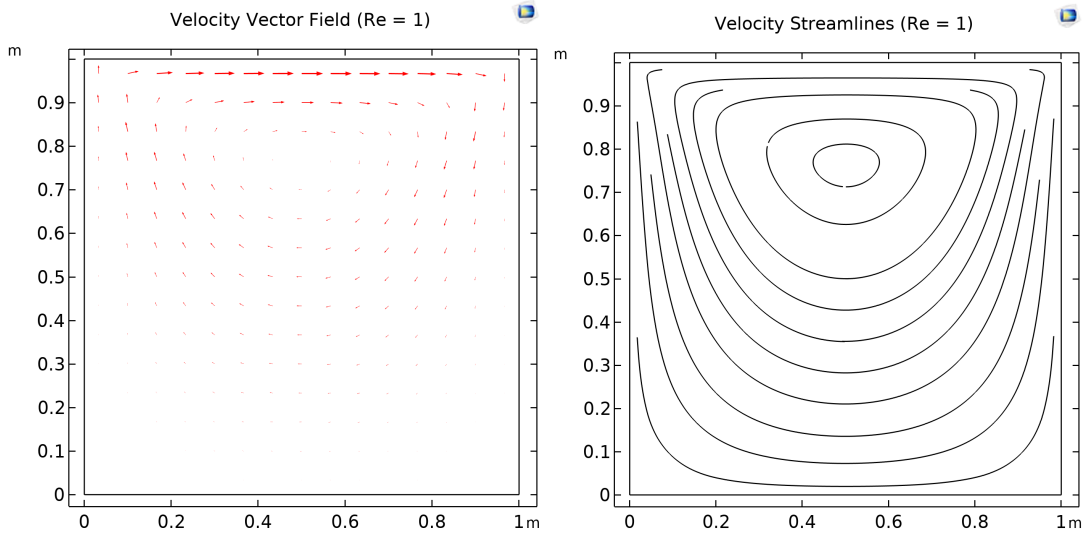


Fig 3. Velocity vector field and streamlines for Re = 1 (side by side)

The velocity vector field and streamlines for Re = 1 are shown in Figure 3. The vector field shows the rightward motion of fluid near the top lid and the return flow along the remaining walls. The streamlines reveal a single, large, primary vortex occupying most of the cavity, with its center located near the center x-position of the domain and slightly shifted in the upward y-direction toward the moving lid. The streamline pattern is practically symmetrical about the vertical centerline, demonstrating the viscosity-dominated flow at low Reynolds number, where asymmetries arising due to inertial effects are suppressed. No secondary vortices are visible, which aligns with the expectation that these features only emerge at higher Reynolds numbers where inertial effects are strong such that recirculation can be sustained in the corners.

3.1.3 Centerline Velocity Profiles

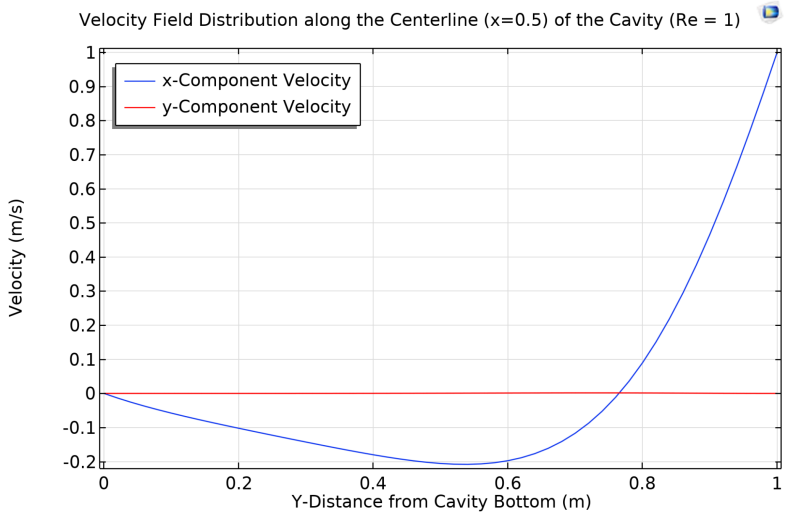


Fig 4. Centerline velocity profiles, u and v, for Re = 1

Figure 4 shows the centerline velocity profiles along x = 0.5 for Re = 1. The x-component of velocity (u) transitions smoothly from a small negative value near the bottom wall, indicating weak flow opposing that of the lid, to the full lid velocity of u = 1 at the top wall. The profile is relatively gradual and monotonic throughout the cavity height, consistent with the smooth, viscous-dominated flow free of sharp velocity gradients in the interior. The y-component of velocity (v) remains close to zero across the entire cavity height at the centerline, which makes sense due to the near-symmetric vortex structure observed in the streamlines, as the vertical velocities along the centerline of a symmetric vortex are minimal.

3.2 High Reynolds Number (Re = 500)

3.2.1 Velocity Magnitude and Vorticity Field

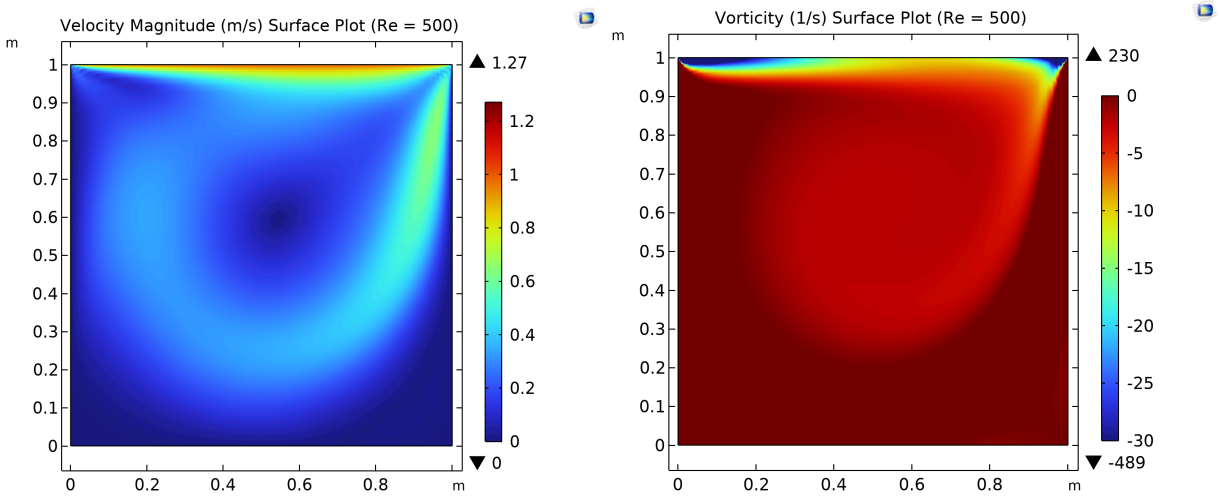


Fig 5. Velocity magnitude surface plot and vorticity surface plot for Re = 500 (side by side)

The velocity magnitude surface plot for Re = 500, shown in Figure 5, displays a significantly different structure compared to the Re = 1 case. High velocity regions are not confined to the top lid but also extend along the right wall, reflecting the stronger inertial transport of momentum along the boundaries before it can be dissipated by viscosity. Additionally, a low-velocity "core" can be seen near the center of the primary vortex. The vorticity field for Re = 500, also shown in Figure 5, shows a much more concentrated distribution compared to Re = 1, with the most intense vorticity seen in thin layers along the top and right walls where there are strong boundary layer effects. There is relatively low magnitude vorticity in the interior of the cavity and the maximum vorticity magnitude (~489 1/s) is much larger than that of the Re = 1 case (~67.8 1/s), demonstrating the steeper velocity gradients that develop in the thinner boundary layers in the high Reynolds number case.

3.2.2 Velocity Vector Field and Streamlines

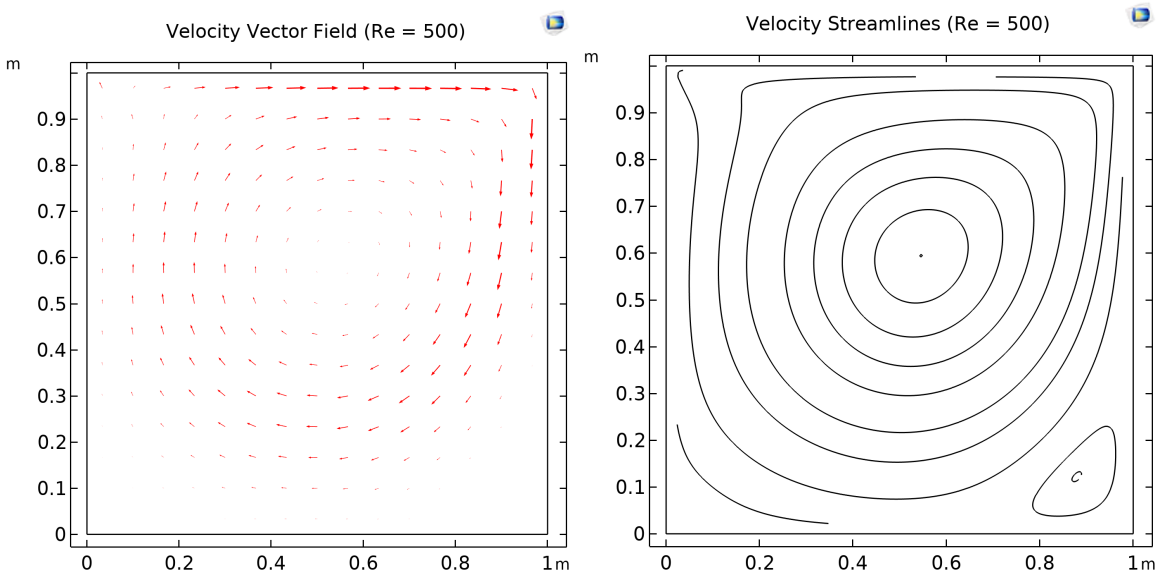


Fig 6. Velocity vector field and streamlines for Re = 500 (side by side)

Figure 6 shows the velocity vector field and streamlines for Re = 500. The vector field reveals stronger, more organized circulation compared to Re = 1, with clear high-velocity flow along the top lid and right wall, and a well-developed return flow along the left and bottom walls. At low Reynolds numbers, the vortex center sits high in the cavity because viscous effects quickly transfer the lid's momentum downward, creating a strong rotational effect concentrated near the top. As Reynolds number increases, inertial effects oppose viscous effects, causing the momentum from the lid to be carried further along the flow direction before diffusing into the cavity interior, shifting the vortex center downward toward the geometric center and slightly in the direction of lid motion. This can be seen in the streamline diagram and characterizes the inertia-dominated flow. Additionally, a secondary vortex can be seen in the lower right corner of the cavity, formed by the separation of the primary vortex flow from the bottom wall. The bottom left corner vortex is not clearly resolved in this simulation, but can be seen when re-running the simulation at higher Reynolds numbers (Re = ~1500).

3.2.3 Centerline Velocity Profiles

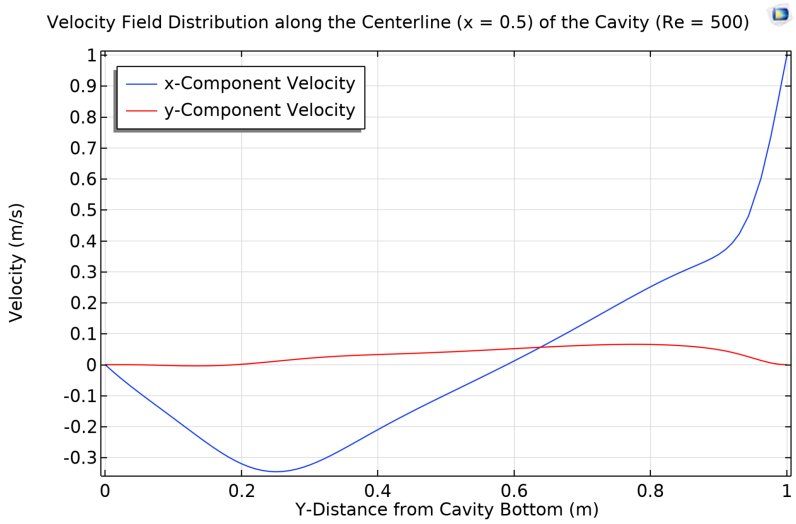


Fig 7. Centerline velocity profiles, u and v, for Re = 500

Figure 7 depicts the centerline velocity profiles along x = 0.5 for Re = 500. The x-component of velocity remains close to zero throughout the lower half of the cavity before increasing steeply toward the lid velocity of u = 1 near the top wall (x = 0.9 to 1 m), reflecting the thin boundary layer and the concentration of momentum transfer in the upper region of the cavity at this higher Reynolds number. A distinct velocity slope can also be observed where the vortex is located (between y = 0.2 and 0.9 m). A negative u-velocity of approximately −0.3 m/s is observed near the bottom wall, indicating a stronger reverse flow compared to Re = 1, driven by the more intense recirculation of the primary vortex. The y-component of velocity remains close to zero across most of the cavity height but shows a small positive peak in the mid-to-upper region (~x = 0.8 m) and a slight negative value near the bottom, showing the effects of the asymmetry of the primary vortex and the secondary corner vortex on the vertical velocity distribution along the centerline.

IV. Conclusion

This study conducted a numerical investigation of two-dimensional lid-driven cavity flow at Reynolds numbers Re = 1 and Re = 500 using COMSOL Multiphysics' Laminar Flow module. The simulations successfully captured the steady-state incompressible flow field for both cases, depicting significant differences in flow structure due to the changing viscous and inertial forces at different Reynolds numbers. At Re = 1, the flow is characterized by a single, symmetric primary vortex centered near the upper middle of the cavity, a smooth vorticity distribution, and gradual centerline velocity profiles. These observations align with the viscosity-dominated creeping flows associated with low Reynolds numbers. At Re = 500, the flow has a more complex structure including a shifted primary vortex center, a secondary vortex in the lower right corner, concentrated vorticity along the lid and slightly along the right wall, and sharper centerline velocity gradients due to the thinner boundary layers that develop at higher Reynolds numbers. Thus, the results agree qualitatively with what is known about low and high Reynolds number flows, validating the computational approach that was used. The mapped mesh (with boundary layer refinement for the Re = 500) proved to be an appropriate strategy for discretization, since it provided enough resolution to capture the key flow features in both the Re = 1 and Re = 500 cases.